

Act III: Casting the Sellers

Production, Costs and Supply Decisions

Module VIII: Seller Sensitivity

The Price Elasticity of Supply



Previously in Marginal Falls

- ◆ Optimal Choices and Demand
- ◆ Cost Curves of a Producer
- ◆ Cost Minimization
- ◆ Optimal combination of inputs
- ◆ Profit Maximization
- ◆ Supply Curve

Next: Seller sensitivity to price changes

Price Elasticity of Supply

Elasticity measures how much one economic variable responds to a change in another variable.

Price elasticity of Supply measures how much quantity supplied changes when price changes.

Price Elasticity of Supply

$$\text{Price Elasticity of Supply} = \frac{\text{Percentage Change in Quantity Supplied}}{\text{Percentage Change in Price}}$$

Mid-Point Formula

Price Elasticity of Supply (E_s) between points (P_1, Q_1) and (P_2, Q_2):

$$E_s = \underbrace{\left(\frac{Q_2 - Q_1}{\frac{Q_1 + Q_2}{2}} \right)}_{\substack{\text{Percentage} \\ \text{Change} \\ \text{in} \\ \text{Quantity} \\ \text{Supplied}}} \div \underbrace{\left(\frac{P_2 - P_1}{\frac{P_1 + P_2}{2}} \right)}_{\substack{\text{Percentage} \\ \text{Change} \\ \text{in} \\ \text{Price}}}$$

Elastic, Inelastic, Unit Elastic

- **Elastic:** quantity responds more than price.

$$E_s > 1$$

- **Inelastic:** quantity responds less than price.

$$E_s < 1$$

$$E_s = \frac{\% \Delta Q}{\% \Delta P}$$

- **Unit elastic:** quantity and price change by the same percentage.

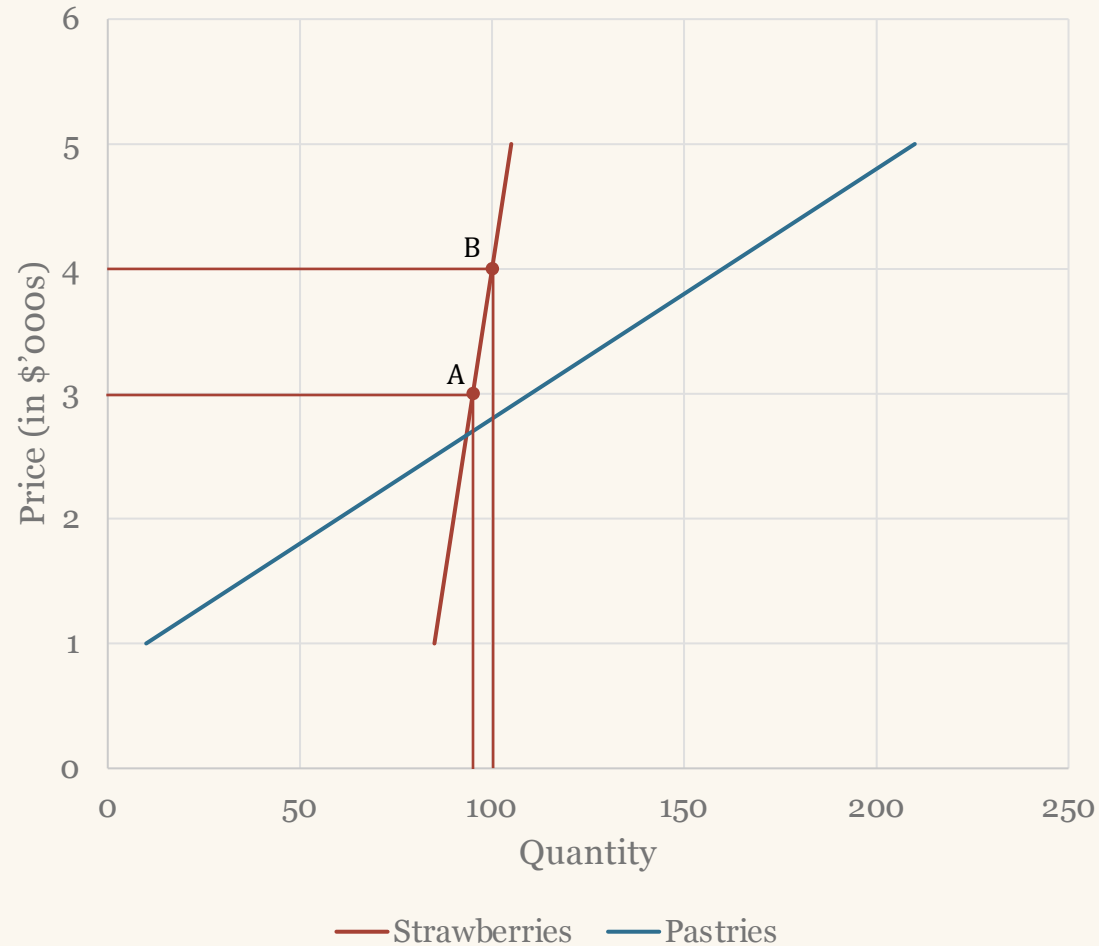
$$E_s = 1$$

$$\% \Delta Q > \% \Delta P$$

$$\% \Delta Q < \% \Delta P$$

$$\% \Delta Q = \% \Delta P$$

Measuring the Sensitivity



Price Elasticity of Supply between A and B for Strawberries:

$$Q_1=95 \quad Q_2=100$$

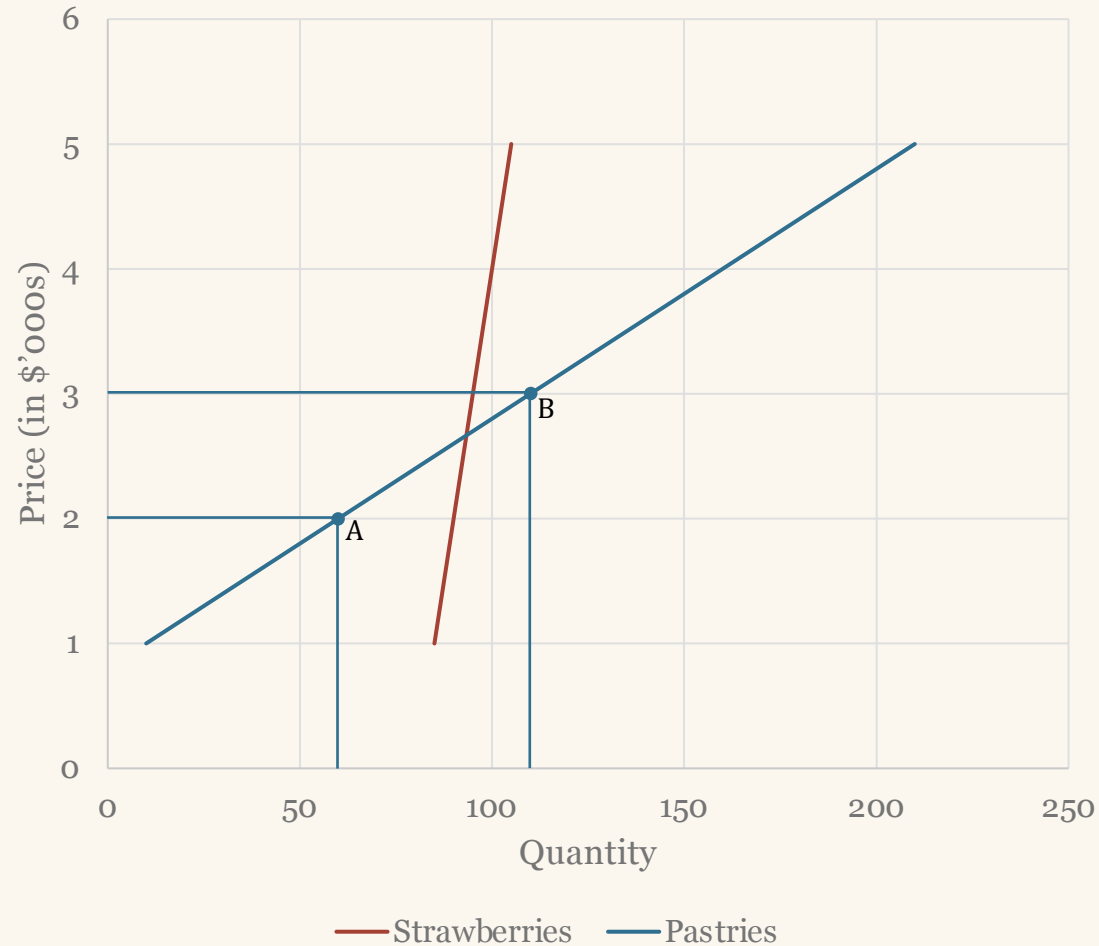
$$P_1=\$3 \quad P_2=\$4$$

Percentage Change in Quantity Supplied:

Percentage Change in Price:

Price Elasticity of Supply:

Measuring the Sensitivity



Price Elasticity of Supply between A and B for Strawberries:

$$Q_1=60 \quad Q_2=110$$

$$P_1=\$2 \quad P_2=\$3$$

Percentage Change in Quantity Supplied:

Percentage Change in Price:

Price Elasticity of Supply: